CLOUD INFRASTRUCTURE & DATA ENGINEERING LAB

Cloud Object Storage Lifecycle Management Lab Report

Complete implementation, automation, verification, and cost-optimization analysis for multi-tier bucket policies (AWS S3, Google Cloud Storage, Azure Blob).

Task 1: Storage Bucket Provisioning & Multi-Application Asset Ingestion

A globally unique storage container was provisioned in AWS S3 (`cloud-storage-lab-app-assets-2026`) to simulate production asset storage across three distinct application domains: media streaming (Spotify), food delivery logistics (Zomato), and fintech payment receipts (Paytm).

```
# Initialize bucket in AWS ap-south-1 (Mumbai) Region aws s3api create-bucket \ --bucket cloud-storage-lab-app-assets-2026 \ --region ap-south-1 \ --create-bucket-configuration LocationConstraint=ap-south-1 # Generate synthetic application payloads echo "Spotify Audio Stream Demo Data" > song.mp3 echo "Zomato Food Order Image Demo Data" > food.jpg echo "Paytm Payment Receipt PDF Demo Data" > bill.pdf # Ingest sample files into bucket root aws s3 cp song.mp3 s3://cloud-storage-lab-app-assets-2026/song.mp3 aws s3 cp food.jpg s3://cloud-storage-lab-app-assets-2026/food.jpg aws s3 cp bill.pdf s3://cloud-storage-lab-app-assets-2026/bill.pdf
```

Target File	App Simulation	Initial Storage Class	Access Requirement
`song.mp3`	Spotify Audio Track	S3 Standard	Low latency, frequent randomized playback
`food.jpg`	Zomato Order Photo	S3 Standard	High immediate access; archival after delivery
`bill.pdf`	Paytm Transaction Receipt	S3 Standard	Short verification window; ephemeral retention

Task 2: Automated Lifecycle Policy Configuration

A declarative lifecycle rule set was enacted to transition `food.jpg` to cold storage (Glacier Flexible Retrieval) after 24 hours, and execute automated permanent deletion on `bill.pdf` after 72 hours.

```
{ "Rules": [ { "ID": "TransitionFoodImageGlacier", "Status": "Enabled", "Filter": { "Prefix": "food.jpg" }, "Transitions": [{ "Days": 1, "StorageClass": "GLACIER" }] }, { "ID": "ExpireBillPDF", "Status": "Enabled", "Filter": { "Prefix": "bill.pdf" }, "Expiration": { "Days": 3 } } ] }
```

The configuration was committed to S3 via the CLI interface:

```
aws s3api put-bucket-lifecycle-configuration \ --bucket cloud-storage-lab-app-assets-2026 \ --lifecycle-configuration file://lifecycle.json
```

Task 3: Operational Monitoring & State Audit (24h to 72h Timeline)

Amazon S3 executes lifecycle state evaluations on a fixed daily schedule (00:00 UTC). The table below outlines observed file states over the evaluation window:

Time Elapsed	Target Object	Storage Class / Status	Observed Behavior & Audit Notes
Hour 0	All 3 files	`STANDARD`	Objects reside in high-availability multi-AZ S3 Standard storage.
Day 1 (24h+)	`food.jpg`	`GLACIER`	Successfully moved to Glacier. Direct GET requests require restore job initialization.
Day 1 (24h+)	`bill.pdf`	`STANDARD`	Unchanged; lifecycle clock running toward day 3 expiration.
Day 3 (72h+)	`bill.pdf`	**Permanently Deleted**	Object expired and purged by S3 daemon. Subsequent access yields HTTP 404.
Continuous	`song.mp3`	`STANDARD`	No lifecycle filter matched; remains permanently available in S3 Standard.

```
# Command used to verify storage class transitions aws s3api list-objects-v2 --bucket cloud-storage-lab-app-assets-2026 \ --query "Contents[].{Object:Key, Class:StorageClass, Modified:LastModified}" --output table
```

Task 4: Financial Impact Assessment & Production Scale Economics

Comparative model simulating a consumer tech application generating **100 TB** of object data monthly in the AWS Mumbai (`ap-south-1`) region:

Architecture Model	Storage Allocation	Unit Cost	Monthly Total
Unmanaged Baseline (No lifecycle)	100 TB in S3 Standard	$0.023 / GB	$2,300.00 / month
Tiered Lifecycle Architecture	10 TB Active S3 Standard 90 TB Cold S3 Glacier Flexible	$0.023 / GB $0.004 / GB	$230.00 $360.00 **$590.00 / month**
Net Realized Savings	**74.3% monthly cloud expense reduction**		**$1,710.00 saved / mo**

Industry Use Cases at Scale

- **Instagram / Reels:** Engagement drops by over 90% after 14 days. Moving video segments older than two weeks into S3 Intelligent-Tiering or Glacier Instant Retrieval saves millions annually without degrading user experience when viewing historical archives.
- **BookMyShow / Ticketing:** Transaction invoices and entry QR passes become obsolete once the show terminates. Automated expiration rules purge transient cache passes after 7 days, while invoices transition to Glacier Deep Archive ($0.00099/GB) for long-term tax audit compliance.

Task 5: Cold Storage & Deletion Policy Model with Production Optimizations

The machine-generated lifecycle specification below transitions audio objects to cold tier at 7 days and permanently deletes them at 30 days:

```
{ "Rules": [ { "ID": "AudioColdTierAndExpirationRule", "Status": "Enabled", "Filter": { "And": { "Prefix": "audio/", "Tags": [{ "Key": "Category", "Value": "PodcastEpisode" }] } }, "Transitions": [ { "Days": 7, "StorageClass": "GLACIER_IR" } ], "Expiration": { "Days": 30 }, "AbortIncompleteMultipartUpload": { "DaysAfterInitiation": 7 } } ] }
```

Production Adjustments & Architectural Rationale

1. **Prefix & Tag Routing over File Extensions:** Cloud object stores do not support wildcard/regex filename filtering (such as `*.mp3`). Production implementations enforce organized folder prefixes (`audio/`) or object metadata tags (`Category=PodcastEpisode`) assigned at upload time.
2. **Glacier Instant Retrieval (GLACIER_IR) Selection:** Standard Glacier requires 1 to 5 hours to reconstitute objects. For audio streaming apps, adopting `GLACIER_IR` ensures millisecond retrieval latency when a user streams an older track, while still slashing storage costs by 68%.
3. **Multipart Cleanup Safeguard:** Added `AbortIncompleteMultipartUpload` to prevent orphaned payload chunks from failed mobile audio uploads from permanently incurring hidden billing costs.